

**School of Computing
Department of Software Engineering**

Course Syllabus

CS210 - Computer Programming 2

**Spring Semester
2026**

Course Title	Computer Programming 2		
Course Code	CS210		
Credits / ECTS	8	Total Hours	180
Contact hours per week	Lecture / Lab		2 / 4
Pre-requisite(s)	CS102		
Semester	Spring	Academic Year	2026
Course leader	Sirojiddin Djurayev		
Email	s.juraev@newuu.uz		
Office Number	218		
Office Hours	Wednesday 11AM-1PM		
Textbook	"Introduction to Java programming" Book by Y. Daniel Liang ISBN-13: 9780138092832 "Head First Java" by Kathy Sierra and Bert Bates		
Reference Books	Harvey Deitel, "Java: How to Program Early Objects (Global Edition)" 11th edition, February 20, 2017		

Course Description (as in the catalog):

A course focused on teaching students the principles of object-oriented programming, including concepts such as classes, objects, inheritance, polymorphism, and encapsulation. Through hands-on coding exercises, students learn how to design and implement robust, modular software solutions using object-oriented techniques.

Course Instructors:

Position	Name	Email	Office	Office hours
Assistant professor	Sirojiddin Djurayev	s.juraev@newuu.uz	218	Wednesday 11AM-1PM
Teaching assistant	Yuliya Durova	y.durova@newuu.uz	310	Monday 2PM-4PM
Teaching assistant	Nargiza Xudoykulova	n.khudaykulova@newuu.uz	208	Thursday 2PM-4PM

Attendance policy:

Attendance is compulsory. We emphasize the importance of the discussions during the instructed hours. A student missing 25% (32 hours or more) of all instructed hours will be forced to withdraw the course (in accordance with the University Academic Regulations).

Course Learning Outcomes (CLOs):

By the end of successful completion of this course, the student will be able to:

No.	Description
CLO1	Understand OOP concepts (e.g., inheritance, polymorphism, abstraction, encapsulation), SOLID principles, and class design patterns.
CLO2	Create custom data types using classes. Make relationships between classes using learned OOP concepts.
CLO3	Understand and utilize features of programming language, such as multiple inheritance, the standard template libraries (STL).
CLO4	Understand concepts of streams, sockets, and concurrent (asynchronous) programming.
CLO5	Design and implement object oriented solutions for small systems involving single/multiple objects. Produce UML object diagrams to solve a particular (software engineering) task.
CLO6	Effectively function as a team member, and develop applications using OOP principles, and effectively communicate (present) their solution.

Contribution level of Course Learning Outcomes(CLOs) to Student/Program Outcomes

Software Engineering

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	Low	Low	Low	High	-
CLO2	Medium	High	Low	Low	Low	High	-
CLO3	Medium	Medium	Low	Low	Low	High	-
CLO4	High	Medium	Low	Low	Low	High	-
CLO5	High	High	High	Low	Low	High	-
CLO6	Low	Medium	High	Medium	High	Medium	-

Cyber Security

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	Low	Medium	Low	Low	-
CLO2	Medium	High	Low	Low	Low	Low	-
CLO3	Medium	Medium	Low	Low	Low	Low	-
CLO4	High	High	Low	Low	Medium	Medium	-
CLO5	High	High	Medium	Low	Low	Medium	-
CLO6	Low	Medium	High	Medium	High	Low	-

AI & Robotics

	S01	S02	S03	S04	S05	S06	S07
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration	Experimental Design and Data Analysis	Lifelong Learning
CLO1	High	Medium	Low	Medium	Low	Medium	-
CLO2	Medium	High	Low	Low	Low	Medium	-
CLO3	Medium	Medium	Low	Low	Low	High	-
CLO4	High	High	Low	Medium	Low	Medium	-
CLO5	High	High	Medium	Low	Low	Medium	-
CLO6	Low	Medium	High	Medium	High	Low	-

Applied Mathematics

	S01	S02	S03	S04	S05
Description	Critical thinking & Problem Solving	Engineering Design and Solution Development	Communication Effectiveness	Ethical and Professional Responsibility	Teamwork and Collaboration
CLO1	High	Medium	Low	Medium	Low
CLO2	Medium	High	Low	Low	Low
CLO3	Medium	Medium	Low	Low	Low
CLO4	High	Medium	Low	Low	Low
CLO5	High	High	Medium	Low	Low
CLO6	Low	Medium	High	Medium	High

Weekly Distribution of Course Topics/Contents

Week	Class Type	Topic	Required Reading	Outcome
1	Lecture / Tutorial	Introduction to Programming and Java. Setting Java Environment. First program using Java.	Chapter 1-2	CLO1, CLO2
2	Lecture / Tutorial	Using the Java Library Random, DateTimeClass etc.. boolean Data Type Control flow Common Mathematical Functions Character String Data Type and Operations Single Dimension Arrays Methods	Chapter 2-8	CLO1, CLO2
3	Lecture / Tutorial	Objects and Classes: Objects vs classes. Class definition, constructors. Static data and methods. Identifying classes. UML class diagrams. Software Engineering Observation: Requirement analysis, Extracting Objects from requirements	Chapter 9	CLO3, CLO4, CLO6
4	Lecture / Tutorial	Object Oriented Thinking: Class design and architecture. Encapsulation/data hiding. Immutability. Access modifiers. Parameterized types (Generics). Coupling and Cohesion. Modularity.	Chapter 10	CLO2, CLO4, CLO6
5	Lecture / Tutorial	Inheritance and interfaces in Java: Overview of Inheritance, inheritance in constructor, Inheriting Data members and methods. Multilevel Inheritance Method overriding. Handling multi level constructors, "super, final" keywords. Overloading Overriding. Abstract Methods and Classes.	Chapter 11	CLO4, CLO6

6	Lecture / Tutorial	Polymorphism. Lifecycle of Objects: Revision	Chapter 11	CLO4, CLO6
7	Lecture / Tutorial	Revision Week	Chapter 12-13	CLO1, CLO5
8	Exam	Midterm Exam	Chapter 1, 2, 3, 4,5, 6, 7, 8, 9 , 10,11	CLO1, CLO3, CLO4, CLO5, CLO6
9	Lecture / Tutorial	SOLID Principles	Chapter 11	CLO4, CLO6
10	Lecture / Tutorial	Exception Handling and Text I/O, Review	Chapter 12-13	CLO1, CLO2, CLO6
11	Lecture / Tutorial	GUI components in Java (Java FX) part 1.	Chapter 14	CLO5, CLO6
12	Lecture / Tutorial	GUI components in Java (Java FX) part 2.	Chapter 14	CLO4, CLO5, CLO6
13	Lecture / Tutorial	JAVA Relationship Database Development, JDBC Data Structures and Algorithms.	Chapter 16	CLO4, CLO5, CLO6
14	Lecture / Tutorial	Java Concurrency (Multithreading), Networking	Chapter 17	CLO3, CLO4
15	Lecture / Tutorial	Revision Week	Chapter 11-17	CLO1, CLO2, CLO3, CLO4, CLO5, CLO6
16	Exam	Final Exam		CLO1, CLO2, CLO3, CLO4, CLO5, CLO6

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Homework/Assignment/Team Project	Every Week	30
Midterm Exam	Week 8	30
Final Exam	Week 16	40
Total		100

Assessment of Course Learning Outcomes:

CLO	Meaning	Evaluation Tool(s) & Weights (tentative)
CLO1	Understand OOP concepts (e.g., inheritance, polymorphism, abstraction, encapsulation), SOLID principles, and class design patterns.	Homework/Assignment/Team Project (30%) Midterm Exam (30%) Final Exam (40%)
CLO2	Create custom data types using classes. Make relationships between classes using learned OOP concepts.	Homework/Assignment/Team Project (30%) Midterm Exam (30%) Final Exam (40%)
CLO3	Understand and utilize features of programming language, such as multiple inheritance, the standard template libraries (STL).	Homework/Assignment/Team Project (30%) Midterm Exam (30%) Final Exam (40%)
CLO4	Understand concepts of streams, sockets, and concurrent (asynchronous) programming.	Homework/Assignment/Team Project (50%) Final Exam (50%)
CLO5	Design and implement object oriented solutions for small systems involving single/multiple objects. Produce UML object diagrams to solve a particular (software engineering) task.	Midterm Exam (50%) Final Exam (50%)
CLO6	Effectively function as a team member, and develop applications using OOP principles, and effectively communicate (present) their solution.	Homework/Assignment/Team Project (100%)

Students' Workload:

Students' workload is distributed as follows:

	Activity	Hrs.	%
Assessment methods and criteria	In-Class Time	84	47 %
	Independent Study	32	18 %
	Assignments	10	6 %
	Examinations	18	10 %
	Group Discussions	25	14 %
	Fieldwork and Practicum	0	0 %
	Workshops and Seminars	0	0 %
	Individual Development	11	6 %
Total		180	100%

Approval			
Course leader - Assistant professor			
Name	Sirojiddin Djurayev		
Signature		Date:	2026-01-23
Department head - Head of Department			
Name	Irfanud Din		
Signature		Date:	2026-01-26
Academic Administration - Head of Department			
Name	Alisher Ikramov		
Signature		Date:	2026-01-26
Department of Quality Assurance and Accreditation - Head of Department			
Name	Rebecca Lee Carter		
Signature		Date:	2026-01-28